

CareFrame

Software Requirements Specification (SRS)

Digital Health Record Management System with AI-Assisted Clinical Summarization

University	KIET Deemed To Be University
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1. Introduction

Purpose. CareFrame is a web-based digital health record system that gives patients and doctors role-specific portals and provides AI-assisted summarization of authorized longitudinal health records.

Scope. The system manages patient profiles, doctor-patient access, encounters, conditions, allergies, medications, observations, medical reports and source documents. It supports PDF/image upload, extraction/OCR, normalization, timelines, trends, conflict detection and evidence-backed AI summaries.

Product principle. CareFrame is an information-assistance system. AI summarizes recorded information and does not autonomously diagnose, prescribe or change treatment.

2. User Classes

User	Responsibilities
Patient	View own records, upload reports, view reports/medications/timeline, receive simplified health summary, manage d
Doctor	View authorized patients, review/update records, inspect reports/trends/conflicts, generate and review AI clinical su
Administrator	Manage users/roles and review audit information.

3. Functional Requirements

- FR-01 Authentication: separate Patient and Doctor login experiences with common secure backend authentication.
- FR-02 Authorization: enforce patient, doctor and administrator permissions server-side.
- FR-03 Patient management: create, retrieve, update and search authorized patient records.
- FR-04 Access management: maintain doctor-patient authorization with active/revoked status.
- FR-05 Medical records: manage encounters, conditions, allergies, medications and observations.
- FR-06 Reports: upload supported PDF/image reports with size/type validation.
- FR-07 Extraction/OCR: extract text and use OCR when usable text is unavailable.
- FR-08 Structured observations: retain value, unit, reference range, date and source document.
- FR-09 Timeline: display clinical events chronologically.
- FR-10 Trends: detect increasing/decreasing/stable/fluctuating/insufficient-data trends only for comparable observations.
- FR-11 Conflicts: flag contradictory information rather than silently resolving it.
- FR-12 AI summary: allow an authorized doctor to generate a structured summary from server-selected context.
- FR-13 Evidence: associate important summary facts with internal source IDs.
- FR-14 Patient summary: provide a simplified summary for the patient.
- FR-15 Audit: record sensitive actions such as login, patient access, record changes, document access and AI generation.

4. Non-Functional Requirements

- Security: bcrypt password hashing, protected APIs, server-side authorization and secure secrets.
- Privacy: synthetic data for development; minimize direct identifiers sent to external AI.
- Reliability: normal EHR functions continue when AI is unavailable.
- Usability: responsive, role-specific interfaces with loading, empty, success and error states.
- Maintainability: business logic in backend services/repositories; UI logic in frontend components/features.
- Traceability: AI statements should link to source records/documents.
- Scalability: documents are stored outside MongoDB and AI context is minimized.

5. System Architecture

Browser → React/Vite frontend → Express/Node.js REST API → MongoDB. Uploaded documents are stored in object storage. The backend performs PDF extraction/OCR and normalization, computes trends/conflicts, and calls the AI provider server-side for structured summaries.

6. Data Requirements

Entity	Key fields
User	name, email, passwordHash, role, timestamps
Patient	patientCode, demographics, contact, address
DoctorPatientAccess	doctorId, patientId, status, access dates
Encounter	patient, doctor, type, reason, notes, occurredAt
Condition	name, status, dates, source
Medication	name, dosage, frequency, status, dates, source
Allergy	substance, reaction, status, source
Observation	name, value, unit, reference range, observedAt, source
Document	file metadata, storage key, checksum, extraction status
MedicalReport	type, document, clinical event date, processing status
AISummary	patient, model, status, summary, source IDs
AuditLog	actor, action, resource, patient, requestId, timestamp

7. AI Safety Requirements

- AI shall summarize existing records rather than act as a medical decision-maker.
- AI shall not prescribe, change dosage or make unsupported definitive diagnoses/treatment decisions.
- AI context shall be built server-side from records the requesting doctor is authorized to access.
- AI output shall be schema-validated before storage/display.
- Conflicts and missing information shall be surfaced explicitly.
- The UI shall state that AI output requires clinician review.

8. Acceptance Criteria

- Patient can log in and access only their own data.
- Doctor can log in and access only authorized patients.
- Supported synthetic reports can be uploaded and processed.
- Structured observations retain units, dates and sources.
- Timeline and trends are displayed correctly.
- Conflicts are flagged.
- Doctor can generate an evidence-backed AI summary.
- AI failure does not break ordinary EHR access.
- Sensitive operations produce audit records.

9. Limitations

This is a project prototype. OCR accuracy, AI factuality, interoperability, consent workflows, regulatory compliance and clinical validation require additional work before real-world clinical deployment. Demonstrations should use synthetic data unless all

applicable requirements for real data have been satisfied.